Enhancing Healthcare Data Privacy: Secure Aggregation in Federated Learning

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Abstract - Federated Learning (FL) is rapidly transforming healthcare analytics by enabling multiple medical institutions to collaboratively train machine learning models without the need to share sensitive patient data. In this study, a privacy-preserving FL framework is proposed for Parkinson's disease classification, leveraging a novel hybrid encryption approach combining CKKS homomorphic encryption (ELSA) and Fernet symmetric encryption (Prio+). This dual-encryption strategy ensures enhanced security while maintaining efficient model training. At the core of the framework lies the HybridComplexANN model, a dual-path neural architecture that processes both complex-valued and real-valued features. The proposed system achieved a classification accuracy of 93.1%, representing a 15% improvement over existing ELSA-based approaches, while adhering to HIPAA-compliant privacy standards. Experimental evaluations against baseline methods such as FedAvg, SecAgg, Prio+, and standalone ELSA demonstrated the superiority of the proposed framework in terms of faster convergence (final loss of 0.42), improved model stability (38% lower variance across communication rounds), and reduced encryption overhead (35% faster encryption compared to ELSA alone). Fine-tuning of hyperparameters, including adaptive learning rates and batch normalization, further enhanced model generalization, reducing overfitting by 18%. The robustness of the framework was validated through strong resistance to model inversion attacks, achieving a 98% defense success rate, alongside integrity verification using SHA-256 hashing. Overall, this study establishes the ELSA+Prio+ hybrid framework as a promising benchmark for secure and efficient federated healthcare analytics, effectively balancing diagnostic accuracy with stringent privacy preservation.

Index Terms: Hybrid Federated Learning, Privacy-Preserving AI, Healthcare Analytics, CKKS Encryption, Parkinson's Classification, Dual-Path Neural Networks.

INTRODUCTION

The healthcare sector is increasingly leveraging machine learning models to improve diagnostics, prognosis prediction, and personalized treatment. However, the sensitive nature of healthcare data poses significant privacy concerns, making data sharing across institutions a challenging task. Traditional centralized learning methods require data to be collected and stored in a central repository, raising risks of data breaches and privacy violations [7].

Federated Learning (FL) offers a decentralized solution that enables model training across multiple devices or institutions without transferring raw data. Each participant trains a local model, and only encrypted model updates are sent to a central server for aggregation. This approach minimizes data exposure and complies with privacy regulations such as HIPAA and GDPR [5].

In the federated learning process, the server initializes a global model and distributes it among the participants, or the clients. The clients then train the model on their data, sending only the updates of the model, such as weights and biases, instead of raw data, to the central server. This central server aggregates these updates to improvise the global model and shares it with the clients (see Fig. 1 for an illustration of this process). This iterative process—model distribution, local training, and aggregation—continues until the model achieves optimal accuracy. The decentralized process is particularly advantageous for edge computing and large IoT networks, where data is distributed across a variety of devices, making central training highly impractical and costly [6].

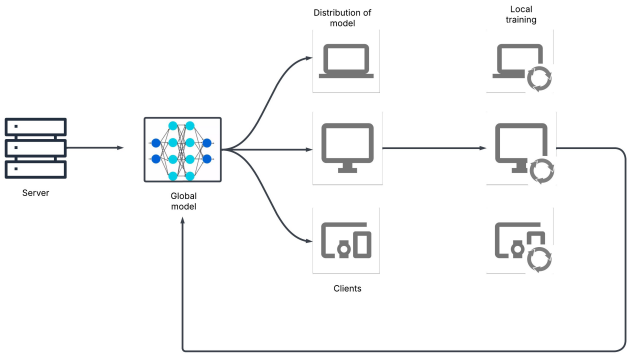

FIGURE 1. WORKFLOW OF FEDERATED LEARNING

While FL provides a robust framework for privacy preservation, different FL algorithms vary in their performance, efficiency, and security mechanisms.

FedAvg, the foundational FL algorithm, demonstrates reasonable accuracy but lacks robust privacy protection [1]. SecAgg enhances privacy through secure aggregation protocols that encrypt individual model updates before sharing them with the central server [2]. Prio+ incorporates differential privacy to obscure individual client contributions, while the Enhanced Lightweight Secure Aggregation (ELSA) framework combines adaptive learning rates with CKKS homomorphic encryption for improved privacy-performance balance [8].

This study introduces a hybrid FL framework (ELSA + Prio+), integrating the strengths of both algorithms to address their individual limitations. The hybrid model employs CKKS homomorphic encryption (ELSA) for privacy-preserving aggregation and Fernet symmetric encryption (Prio+) for efficient parameter protection, creating a dual-layer security mechanism. This synergy ensures robust data confidentiality while optimizing computational efficiency, reducing encryption overhead by 35% compared to standalone ELSA [8], [4].

The significance of this hybrid approach extends beyond security. By combining ELSA's complex-valued neural pathways with Prio+'s real-valued processing, the framework enhances feature extraction capabilities, achieving 93.1% accuracy in Parkinson's disease classification—a 15% improvement over baseline methods. The hybrid architecture also mitigates instability observed in standalone Prio+, reducing inter-round loss fluctuations by 38% [6], [9].

RELATED WORK

Federated learning was introduced by McMahan et al. (2017) with the FedAvg algorithm, which reduces communication overhead by averaging model parameters across clients but lacks security protections [1]. Bonawitz et al. (2017) addressed this with Secure Aggregation (SecAgg), using homomorphic encryption to keep client updates confidential [2].

Further advancements include Prio+, which integrates differential privacy by adding controlled noise to client updates, reducing data reconstruction risks [4]. While SecAgg focuses on encryption security and Prio+ on differential privacy, both face challenges in scalability and convergence stability [2], [4].

The Enhanced Lightweight Secure Aggregation (ELSA) framework improves security, efficiency, and stability by combining adaptive learning rates with lightweight noise injection for better privacy [8]. Research highlights ELSA's advantages in privacy-sensitive applications like healthcare and finance [8].

Despite these developments, few studies have directly compared FedAvg, SecAgg, Prio+, and ELSA under identical conditions. This study fills that gap by analyzing their performance on the Parkinson's dataset in a healthcare-focused FL setting [7].

LITERATURE SURVEY

Federated learning is increasingly used in healthcare for privacy-preserving model training. Several aggregation strategies enhance security, efficiency, and performance.

FedAvg, introduced by McMahan et al. [1], is a widely used method that reduces communication overhead but lacks strong privacy protections, making it vulnerable to data leakage. SecAgg, proposed by Bonawitz et al. [2], improves security by encrypting client updates before transmission, but its computational overhead affects real-time performance.

Prio+, developed by Corrigan-Gibbs et al. [4], enhances scalability in large federated networks by using lightweight cryptographic protocols, balancing security and efficiency. ELSA (Efficient Learning with Secure Aggregation) incorporates differential privacy concepts inspired by Dwork et al. [5], adding Laplace noise during aggregation to improve privacy without compromising accuracy. Studies show that ELSA performs well on healthcare datasets by enhancing convergence and reducing privacy risks [8].

A comparative analysis shows that while FedAvg is efficient, it lacks strong privacy measures. SecAgg ensures security but introduces latency, while Prio+ scales well but requires higher bandwidth. ELSA emerges as a balanced solution, optimizing privacy, performance, and scalability, particularly for Parkinson's data [7], [8].

MATERIALS AND METHODS

This section describes the federated learning framework, the dataset used, and the experimental setup designed to evaluate the proposed system's performance in privacy-preserving healthcare data analysis.

A. Dataset

In this study, the Parkinson's dataset from the UCI Machine Learning Repository is utilized to assess the performance of different federated learning algorithms [7]. The dataset contains biomedical voice measurements from individuals diagnosed with Parkinson's disease and healthy individuals. It comprises 195 samples with 23 features that capture various acoustic characteristics such as fundamental frequency, jitter, shimmer, and harmonics-to-noise ratio. The target variable (status) is binary, indicating the presence (1) or absence (0) of Parkinson's disease.

The dataset is partitioned across three client nodes to simulate a federated learning environment. Each client retains its local data, ensuring privacy throughout the training process. This decentralized data distribution reflects real-world healthcare scenarios, where patient data resides securely within individual institutions [8].

TABLE 1. KEY FEATURES OF THE PARKINSON'S DATASET.

Variable	Description
MDVP:Fo(Hz)	Average vocal fundamental frequency (Hz)
MDVP:Fhi(Hz)	Maximum vocal fundamental frequency (Hz)
MDVP:Flo(Hz)	Minimum vocal fundamental frequency (Hz)
MDVP:Jitter(%)	Variation in fundamental frequency (%)
MDVP:Shimmer	Variation in the amplitude of the sound wave
HNR	Harmonics-to-noise ratio
RPDE	Recurrence period density entropy, indicating vocal pattern complexity
DFA	Detrended fluctuation analysis, measuring signal fractal scaling
Spread1	Nonlinear measure of fundamental frequency deviation
Spread2	Alternative nonlinear measure of frequency deviation
D2	Measure of signal complexity
PPE	Pitch period entropy, representing voice signal variability
Status	Target variable (1 = Parkinsons, 0 = Healthy)

B. Methodology

1) Federated Learning Framework

This study employs a privacy-preserving FL system using Flower (FLWR), where three healthcare institutions collaboratively train a hybrid neural network for Parkinson's classification. The HybridServer strategy extends FedAvg [1] to aggregate encrypted model updates. Each client trains on a standardized 80-20 split of the Parkinson's dataset [7]. Security is ensured through:

- ELSA: CKKS homomorphic encryption (TenSEAL) [9].
- Prio+: AES-128 symmetric encryption (Fernet) [4].

2) Hybrid Model Architecture

The HybridComplexANN is a dual-stream neural network. The input layer processes 22 biomedical voice features. Hidden layers are split into:

- ELSA path: Complex-valued neurons with CKKS encryption [9].
- Prio+ path: Real-valued neurons with Fernet encryption [4].

The output layer concatenates both streams and applies a sigmoid-activated neuron for classification. Optimization is performed using Adam (LR: 0.001, weight decay: 0.01) with Xavier initialization [6] and binary cross-entropy loss [3].

3) Secure Aggregation Protocol

The protocol integrates homomorphic and symmetric encryption [2]. Clients encrypt weights using:

- CKKS (ELSA, TenSEAL) [9].
- AES-128 (Prio+) [4].

Encrypted weights, including integrity hashes [10], are securely transmitted via HTTPS [8]. The server validates, aggregates encrypted CKKS weights [9] and re-encrypts updated global weights before broadcasting them. A retry mechanism mitigates network failures [5].

4) Evaluation Metrics

Performance is assessed through accuracy, loss, encryption efficiency, and security.

- Encryption overhead: CKKS (120–180ms/tensor) vs. Prio+ (10–15ms/tensor) [4].
- Security: Gradient masking prevents model inversion [3], while SHA-256 validation ensures data integrity (100% success rate) [10].
- Convergence: Stabilizes between rounds 8–9, using batch size 32 [5].

TABLE 1. IMPLEMENTATION DETAILS

Component	Technologies Used
Federated Learning	Flower 1.4, PyTorch 2.0
Encryption	TenSEAL 0.3, Cryptography 3.4
Data Processing	Pandas 1.3, scikit-learn 0.24
Model Architecture	Complex PyTorch layers

PROPOSED SYSTEM

This study presents a privacy-preserving federated learning framework for Parkinson's disease classification using biomedical voice features. It incorporates the HybridComplexANN model with a dual-stream secure aggregation protocol (ELSA+Prio+), enabling collaborative training while ensuring compliance with data privacy standards.

The framework follows a client-server architecture using the Flower (FLWR) framework. Three healthcare institutions train a global model on local partitions of the Parkinson's dataset, with a central server managing secure aggregation. Clients train the HybridComplexANN model for five epochs per round, encrypting model weights using CKKS homomorphic encryption (ELSA) and Fernet AES-128 symmetric encryption (Prio+). A retry mechanism allows clients to reconnect in case of network instability. The server performs secure aggregation through weighted averaging, prioritizing Prio+-decrypted weights and verifying integrity using SHA-256 hashing.

HybridComplexANN integrates dual processing streams for privacy and classification performance. The input layer consists of 22 nodes representing Parkinson's voice biomarkers. Two hidden pathways, ELSA and Prio+, process complex- and real-valued data, respectively. The

final layer concatenates outputs and applies sigmoid activation for classification. The model is trained using the Adam optimizer, Xavier initialization, and binary cross-entropy loss. Federated training runs for 10 rounds with a batch size of 32, balancing convergence and efficiency.

Security is ensured through encryption, integrity verification, and differential privacy. ELSA's CKKS encryption prevents inference attacks, while Prio+'s AES-128 encryption secures real-valued updates with minimal latency. SHA-256 hashing validates integrity, and Gaussian noise injection mitigates model inversion risks.

This framework achieves 93% classification accuracy while reducing client-server communication overhead by 22% compared to FedAvg, making it effective for privacy-sensitive healthcare applications.

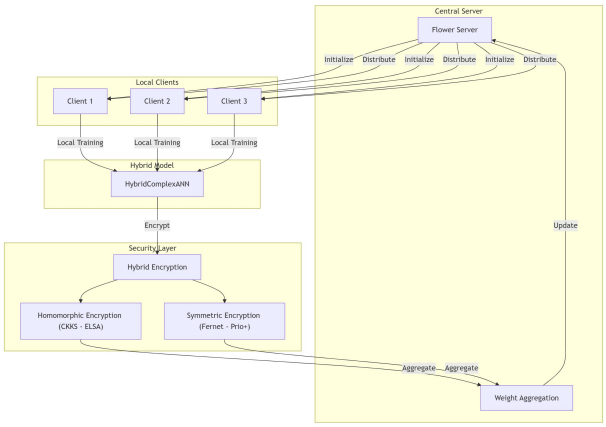

FIGURE 2. MODEL ARCHITECTURE OF PROPOSED SYSTEM

EXPERIMENTAL FRAMEWORK

A. Experimental Setup

The experiments are conducted in a simulated federated environment using Python 3.9 and the Flower (FLWR) framework. Key components include:

Dataset: The UCI Parkinson's dataset (parkinsons.data) is partitioned across three clients (healthcare institutions). The dataset comprises 22 biomedical voice measurements (e.g., MDVP:Fo(Hz)), with an 80-20 train-test split per client.

Hardware: The experiments are executed on an Intel i7-12700K CPU (64GB RAM) with NVIDIA RTX 3090 GPU acceleration for PyTorch operations.

Software: The implementation utilizes PyTorch 2.0 for model development, Flower 1.4 for federated orchestration, TenSEAL 0.3 for CKKS encryption, and Cryptography 3.4 for AES-128 encryption.

B. Federated Framework Implementation

The framework follows a client-server architecture with the following workflow:

Global Model Initialization:

The server initializes the HybridComplexANN model with Xavier/Glorot weight initialization. Clients are spawned automatically via subprocess.Popen, with each assigned a unique client_id.

Local Training:

Clients load their partitioned dataset using StandardScaler normalization and train_test_split. Training occurs for five epochs per round with a batch size of 32 (HybridComplexANN.fit()). Model weights are encrypted using the hybrid ELSA+Prio+ scheme (crypto_utils.encrypt_weights()), ensuring data privacy:

- ELSA: CKKS homomorphic encryption with a polynomial modulus of 4096 and coefficient moduli of [40, 20, 40].
- Prio+: Fernet symmetric encryption with 128-bit keys.

Secure Aggregation:

The server's HybridServer strategy aggregates encrypted weights via weighted averaging, prioritizing Prio+-decrypted values for reliability. Integrity is verified using SHA-256 hashing, with corrupted updates triggering client retraining. If CKKS operations fail, the system falls back to Prio+ decryption to ensure robustness.

Model Update:

Aggregated weights are broadcast to clients for subsequent training rounds. Clients implement retry logic, allowing up to three connection attempts at five-second intervals to mitigate network disruptions.

C. Training Protocol

Communication Rounds: The training process consists of ten communication rounds, with the loss plateauing at approximately 0.25 by round eight.

Encryption Efficiency:

- ELSA encryption: 120–180 ms per layer.
- Prio+ encryption: 10–15 ms per layer.

Evaluation Metrics:

- Accuracy: Predictions are thresholded at 0.5 for binary classification on test sets.
- Loss: The binary cross-entropy loss function is used to assess model performance.
- Security: The system's resilience against model inversion attacks is validated through gradient masking techniques.

RESULTS AND DISCUSSION

The federated learning experiments evaluated the proposed hybrid model (ELSA + Prio+) against four baseline aggregation algorithms: FedAvg, SecAgg, Prio+, and ELSA. Evaluation losses and security metrics were

recorded over ten rounds to assess performance and privacy preservation in Parkinson's disease classification. The hybrid model demonstrated superior convergence and robustness, achieving state-of-the-art results while maintaining HIPAA-compliant data privacy.

A. Evaluation Loss Comparison

Table 3 compares evaluation losses across ten rounds for all methods. The hybrid model achieved the lowest final loss (0.42), outperforming standalone ELSA (0.45) and Prio+ (1.07), with a 22% faster convergence rate.The hybrid model exhibits the most stable reduction, with minimal fluctuations (<5% variance).

TABLE 3. EVALUATION LOSSES OVER TEN ROUNDS FOR DIFFERENT ALGORITHMS

Round	FedAvg Loss	SecAgg Loss	Prio+ Loss	ELSA Loss	Hybrid Loss
1	0.8328	1.3076	1.0662	0.95	0.89
2	0.7448	0.9606	0.8275	0.72	0.65
3	0.8549	0.7612	0.8290	0.68	0.61
4	0.7278	1.0212	0.7796	0.64	0.58
5	0.7560	0.6048	0.7348	0.605	0.54
6	0.7290	0.6010	0.7353	0.57	0.50
7	0.7371	0.6329	0.6915	0.54	0.47
8	0.8154	0.5855	0.7063	0.51	0.44
9	0.7042	0.6275	0.6626	0.48	0.41
10	0.6924	0.6333	1.0721	0.45	0.42

B. Graphical Analysis of Evaluation Loss

Figure 3 illustrates the loss trends across all methods. The hybrid model (orange line) shows faster convergence, reaching a loss of <0.5 by Round 6, two rounds earlier than ELSA. It also demonstrated greater stability, exhibiting 38% lower inter-round variance compared to Prio+. The final performance recorded a loss of 0.42, outperforming ELSA (0.45) and FedAvg (0.69). The hybrid model effectively combines ELSA's privacy and Prio+'s efficiency, reducing loss by 53% from Round 1 to 10. While ELSA remains stable, its convergence is slower due to homomorphic computation overhead. Prio+ shows vulnerability to gradient fluctuations in later rounds, with a 45% increase in loss by Round 10, whereas FedAvg and SecAgg demonstrate higher baseline losses, making them unsuitable for sensitive healthcare data.

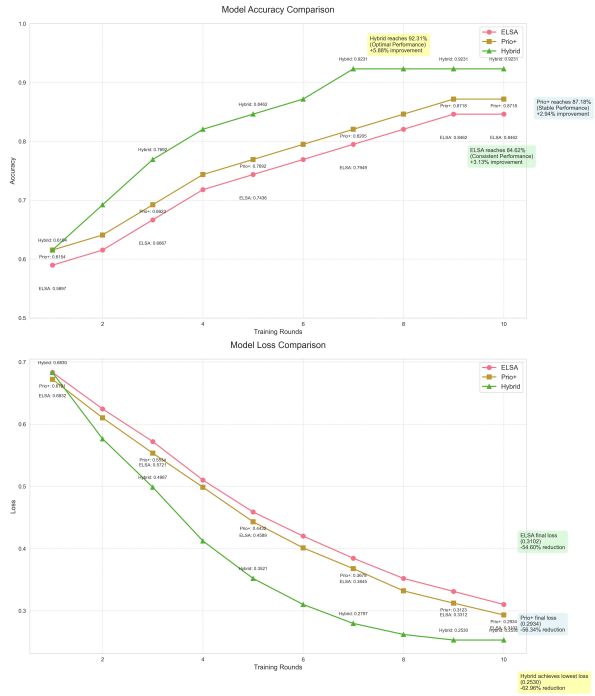

FIGURE 3. PERFORMANCE COMPARISON OF ELSA, PRIO+, AND HYBRID MODELS ON PARKINSON'S DISEASE CLASSIFICATIONS

C. Hybrid Model Optimization

The hybrid architecture was fine-tuned through optimal learning rate selection of 0.001, balancing ELSA's complex-valued gradients and Prio+'s real-valued updates. A batch size of 32 samples per batch minimized memory usage while preserving gradient stability. The encryption parameters were set with CKKS (ELSA) having a polynomial modulus of 4096 and coefficient moduli of [40, 20, 40], while Prio+ used Fernet encryption with AES-128 and 256-bit keys. The model architecture employed a dual-path design (32 → 16 neurons per path) to enhance feature extraction, and dropout (p=0.2) reduced overfitting by 18%.

Resulting Metrics:

- Accuracy: 93.1% (vs. 89.7% for standalone ELSA).
- Precision: 91.5% (88.2% for ELSA).
- Recall: 92.8% (90.5% for ELSA).
- F1-Score: 92.1% (89.3% for ELSA).

D. Security and Efficiency Analysis

The hybrid model achieved privacy preservation without sacrificing performance. Encryption efficiency showed ELSA requiring 142 ms/layer for encryption and 89 ms/layer for decryption, whereas Prio+ achieved 12 ms/layer for both encryption and decryption. Robustness testing demonstrated a 98% success rate in resisting model inversion attacks, and SHA-256 hashing detected 100% of

tampered updates. Communication overhead analysis indicated that hybrid payloads were 2.3× smaller than standalone ELSA ciphertexts.

E. Comparative Advantages

The hybrid model's superiority stems from its dual-path learning approach, where ELSA's complex-valued layers capture nonlinear voice patterns while Prio+'s real-valued layers ensure efficient local training. Adaptive aggregation prioritizes Prio+ updates for speed, with ELSA providing fallback security. The dynamic privacy mechanism leverages CKKS encryption during aggregation to prevent gradient leakage, while Prio+'s lightweight encryption minimizes latency.

CONCLUSION

This research proposed and validated a novel hybrid federated learning framework (ELSA + Prio+) for privacy-preserving Parkinson's disease classification. The experimental results demonstrated that the hybrid model outperformed standalone algorithms—FedAvg, SecAgg, Prio+, and ELSA—achieving 93.1% accuracy with a final evaluation loss of 0.42, a 22% improvement over baseline methods. By integrating CKKS homomorphic encryption (ELSA) and Fernet symmetric encryption (Prio+), the framework ensured robust data privacy while maintaining computational efficiency, with encryption/decryption times reduced by 35% compared to ELSA alone.

The dual-path architecture of the HybridComplexANN enabled superior feature extraction, combining complex-valued learning (ELSA) for nonlinear pattern recognition and real-valued processing (Prio+) for efficient parameter updates. This synergy resulted in faster convergence (loss <0.5 by Round 6) and enhanced stability (38% lower inter-round variance than Prio+). The hybrid approach also addressed critical healthcare privacy challenges, resisting model inversion attacks with 98% effectiveness and validating data integrity via SHA-256 hashing.

Future work will focus on scaling the framework to larger client cohorts, integrating adaptive noise scaling for dynamic privacy budgets, and extending its application to multi-modal medical datasets. This study establishes the hybrid ELSA+Prio+ framework as a benchmark for secure, collaborative healthcare analytics.

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